1. INTRODUCTION TO ENVIRONMENTAL ECONOMICS

I) ORIGINS OF ENVIRONMENTAL ECONOMICS

Environmental economics developed in its present form in the 1960s as a result of the intensification of pollution and the heightened awareness among the general public in Western countries about the environment and its importance to our existence. Economists became aware that, for economic growth to be indefinitely sustainable, the economic system needs to take into account the uses of the environment that we have already mentioned, so that natural resources are not depleted and so that the environment is not overused as a waste sink.

Environmental economists view the environment as a form of natural capital which performs life support, amenity, and other functions that cannot be supplied by man-made capital. This stock of natural capital includes natural resources plus ecological systems, land, biodiversity, and other attributes.

The growth of environmental economics in the 1970s was initially within the neoclassical paradigm. In general, this approach to the environment is concerned with issues of market failure, inappropriate resource allocation, and how to manage public goods. There was little concern for the underlying relationships between the economy and the environment. Concerns about the limits of this approach to environmental economics led some environmental economists to develop what is now referred to as ecological economics. Ecological economics views the relationship of the economy and the environment as central. Thus, any analysis places economic activity within the environment. This distinction is best illustrated with reference to debates concerning sustainable development and the difference between weak and strong sustainability. Ecological economics supports the notion of strong sustainability. This view of sustainability assumes that not all forms of capital (ie human and natural) are perfectly substitutable.

II) MEANING OF ENVIRONMENTAL ECONOMICS

Economics is no longer merely a science of production and distribution; it has to take into account the ecological repercussions of economic activities that could affect both production and distribution. It means that economics as a subject cannot exist in isolation, it cannot even be a mere study of how goods and services are produced, but at the same time it has to take into consideration the impacts of the use of resources on the environment. The impacts may be

in the form of externality, pollution, exhaustion, etc. Any study on the economic content of production, distribution, development, etc., cannot be completed without touching upon the environmental aspects like externality, pollution, damage, exhaustion, depletion etc.

Environmental economics can therefore be defined as that part of economics which deals with interrelationship between environment and economic development and studied the ways and means by which the former is not impaired nor the latter impeded It is thus a branch of economics which discusses about the impacts of interaction between men

and nature and finds human solutions to maintain harmony between men and nature. Environmental economics teaches us how to promote economic growth of nations with least environmental damage. Classical and neoclassical school of thoughts underestimated the environmental issues of production and consumption, since they considered these issues merely as social issues. When the environmental goods get transferred into economic goods, the problems of environmental damage crop up, and therefore the

need to interact with economic principles.

III) NATURE OF ENVIRONMENTAL ECONOMICS

1. Positive and Normative aspects:

Environmental economics is an application of scientific theories and general application of welfare economics. When we study the cause-and-effect relationship, it covers the positive aspect. For example, the laws of thermodynamics are equally applicable to economic process. If the problem is related to policy measures, then it is considered as normative aspect. Therefore, environmental economics is a normative science because it prescribes the goals of environmental policy.

2. A Study of Micro and Macro Aspects:

We generally observe crowded market places, industrial units, and even residential areas in a city like, Delhi. We do not get enough fresh air at these places. Its solution lies in micro level planning. On the other-hand, when the pollution problem is related to the economy as a whole such as rise in temperature, then it is related to macro aspect of environmental planning.

Environmental economics draws more from microeconomics than from macroeconomics. It focuses primarily on how and why people make decisions that have consequences for the natural environment. It is concerned also with how economic institutions and policies can be changed to bring these environmental impacts more into balance with human desires and the needs of the ecosystem.

3. Environmental Pollution as an Economic Problem:

Environmental pollution is an economic problem because it requires us to make choices and to resolve conflicts of interests. It is an economic problem because the means by which pollution can be reduced are themselves resources using. Further, it also reduces the value of some resources that society has at its disposal. Economic growth can affect environmental quality under different situations. Environmental quality can increase with economic growth. Thus, increased incomes, for example, provide the resources for public services.

4. As a Social Science:

Environmental economics deals with economic and managerial aspects of pollution and natural resources. It interacts between human beings and their physical surroundings. It studies the impact of pollution on human beings and suggests rational utilization of resources in a proper way so that there may be an increase in social welfare or minimization of social costs.

IV) SCOPE OF ENVIRONMENTAL ECONOMICS

Environmental economics is considered both a positive and a normative science. Therefore, it has wide scope.

1. Economy-environment analysis:

Environmental economics is primarily concerned with the impact of economic activities on environment and its implications for the individual firm, industry and the economy as a whole. Economists have formulated economy-environment models to explain the various economic activities and their external effects. For example, the Material Balance Model and the Leontief Abatement Model explain these externalities.

2. Eco-development:

The main objective of environmental economics is to maintain a balance between economic development and environmental quality. In order to achieve it, environmental economists have to explore the various socioeconomic possibilities to reduce pollution and uplift the standard of living of the people. This objective gained momentum after the publication of the Report on Limits to Growth

3. Welfare approach:

Environmental economics has emerged as a discipline to tackle environmental problems from an economic welfare framework. The welfare framework covers scarce resources and market failures due to property rights and ethical aspects of different problems of pollution. Thus, it suggests the best possible means to tackle the environmental problems.

4. Environmental values:

Environmental issues are about resources. The neo-classical economists have analysed the use of various resources like fisheries, forests, fossil fuels and water in a rational manner and with environmental values. In fact, environmental values are economic values. It is important for the society to conserve its limited resources in the interest of economic efficiency and welfare.

5. Clean Technology:

Presently environmental pollution is caused by misuse of existing technology and failure to develop better one. Environmental economists are in favour of appropriate and clean technologies which provide the most rational use of natural resources and energy and to protect the environment.

6. Conservation Policy:

The longstanding foundation of environmental economics lies in conservation economics which tends to emphasise the impact of economic activities on demand for productive resources and energy resources. It suggests the optimal strategy in the utilization of natural resources in a rational manner.

7. Multi-disciplinary base:

Environmental economics is inherently a multi-disciplinary subject. It consists of an integration of many varied disciplines such as biology, ecology, physical sciences, ethics and main stream economics. Therefore, it has wide scope.

V) SIGNIFICANCE OF ENVIRONMENTAL ECONOMICS

1. Environmental economics will help you understand some important and controversial issues such as climate change policy, nuclear power, recycling policy, and traffic congestion charging. This is an exciting field of economics to study and very much at the heart of many public debates and controversies.

2. Environmental economics is primarily concerned with the impact of economic activities on environment and its implications for the individual firm, industry and the economy as a whole. Economists have formulated economy-environment models to explain the various economic activities and their external effects.

3. The main significance of environmental economics is to maintain a balance between economic development and environmental quality. In order to achieve it, environmental economists are exploring the various socio-economic possibilities to reduce pollution and uplift the standard of living of the people.

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VI) ECONOMY AND THE ENVIRONMENT

Man cannot exist in isolation. Man's life is interconnected with various other living and non-living things. His life also depends on social, political, economic, ethical, philosophical and other aspects of social system. In fact, the life of human beings is shaped by his living environment. What exactly is living environment? Environment means ―all the conditions, circumstances, and influences surrounding and affecting the development of an organism or group of organisms. It also means that the complex of physical, chemical and biotic factors that act upon an organism or an ecological community and ultimately determine its form and survival.

Environment, environmentalists, environmentalism etc., are the common words used in our ordinary life in recent years. Environmentalists are those who love and care for environment, who realize that any damage to the environment will affect the life of living things. Environmental concern of environmentalists and fundamental environmentalists are different. The former upholds and tries to popularize the need for

environmental education. But the latter embraces environment in its virgin form and any intervention in the ecological balance of the environment mars the very survival of living things. Therefore, fundamental environmentalists are always treated as anti-developmentalist. But the works of such persons are always appreciated by the people at large.

The words Ecology and Economics stem from the same Greek root Oikos which means habitation. Ecology is the study of the relationship or interdependence between living organisms and their environment. Hence in Greek root, Ecology deals with the household and nature, while Economics deal with the household of man. An ecological balance exists in the society in which all the living things live harmoniously. But the problem is that man in his aspiration for better living has upset the ecological balance thereby endangering nature as well as himself. Quite often we find that there is a conflict between Economy and Ecology. Ecology studies harmony between nature and man, whereas Economics spells out the disharmony between man and nature. The disharmony arises as a result of the incompatibility of the basic ecological principle of stability as a precondition for the sustainability of ecological system and the economic principles of business profitability.

Environmental economics attempts to study the inter relationship between economic and environment. Economic as a subject cannot exist in isolation, it cannot even be a mere study of how good and services are produced, but at the same time it has to take into consideration the impacts of the use of resources on the environment. The impact may be in the form of externality, pollution, exhaustion, etc. Any study on the economic content of production, distribution, development, etc., cannot be completed without touching upon the environmental aspects like externality, pollution, damage, exhaustion, depletion etc. environmental economics can therefore be defined as that part of economics which deals with interrelationship between environment and economic development and studies the ways and means by which the former is not impaired nor the letter impeded. Environmental economics teaches us how to promote economic growth of nations with least environmental damage. When the environmental goods get transferred into economic goods, the problems of environmental damage crop up and therefore the need to interact with economic principles. One must begin by recognizing the threefold connection between the environment, human society and its economy.

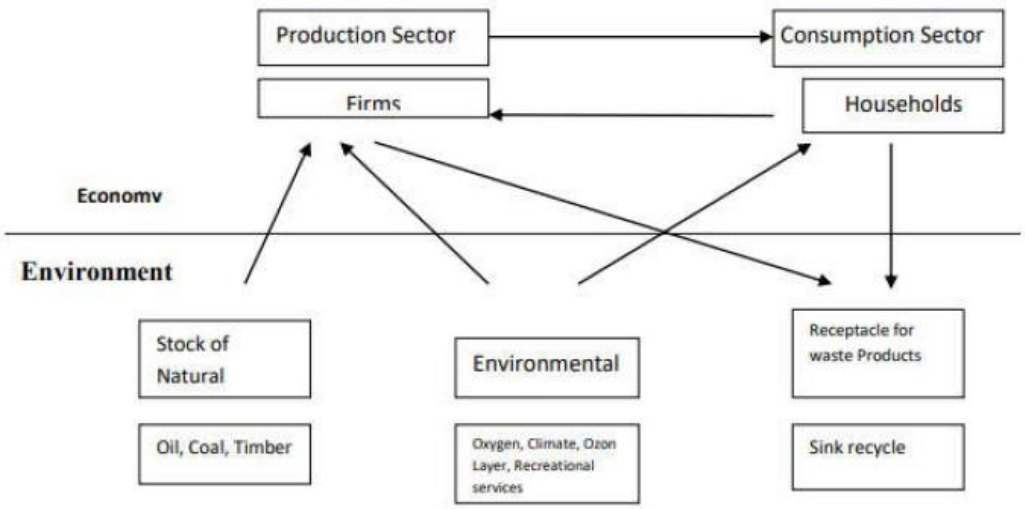
First, the environment provides the economy with raw materials which are transformed into consumer products through the production process. These raw materials include energy, which is itself a consumer product as well as an intermediate that drives this transformation.

Second, the environment provides services which are used directly by consumers. These may be critical life-support services such as the oxygen in the air that we breath or the water that we drink. They may be aesthetic or recreational services that we may derive pleasure from, such as rambling in the forest or boating on the river.

Finally, there is a less recognized but vital service that the environment provides to the economy. It acts as a receptacle or a sink for all the waste products that are the result of the

process of production and consumption. The environment is not a passive sink, it acts upon the waste products to clean up the environment and recycle the waste into material that can be used again.

These inter linkages are given in the following diagram:



All problems relating to the degradation of the environment relates somehow to an interference that occurs in this relationship, that hinders the delivery of these good and services that are provided directly or indirectly by the environment to the economy. It may involve the slowing down or a complete breakdown of the natural clean-up process. It is this interference or obstacle that lies at the root of all environmental problems.

Let us take a simple example, if a factory produces some good, it also produces smoke. The amount of good it produces is decided by the economy. This in turn decides the amount of smoke that will be belched out and the damage to human health it will cause. If your neighbour plays his music system too loudly, your mental peace is disturbed. Thus, all problems of environmental pollution or degradation occur as a byproduct of our activities related to production or consumption. It is therefore, important to understand the economic forces that derive production and consumption such as the formation of market prices and the optimal allocation of inputs.

VII) ENVIRONMENTAL KUZNETS CURVE

The Environmental Kuznets curve relationship between environment and development takes its inspirations from the income distribution theory developed by Simon Kuznets in 1955. In his study, Kuznets found an inverted U-shaped relationship between the indicators of income inequality and the level development as measured by per capita income.

The income inequality increases along the path of economic development in the early phase, declares in the later phase. Gene Grossman and Alan Krueger in their studies of the relationship between the environment degradation and economic development found a

similar inverted U-shaped relationship. This inverted U-shaped relationship between the environmental degradation and economic development is known as Environmental Kuznets Curve (EKC). The EKC hypothesis expresses the most likely relationship between the environment and economic development. It states that the environmental degradation is low when the level of economic development is low. The environmental degradation increases with economic development in the early phase but it comes down at the later stage of development. That is, in the initial stage of development, environmental degradation increases but eventually declines at certain threshold level of income.

Definition: The environmental Kuznets curve suggests that economic development initially leads to a deterioration in the environment, but after a certain level of economic growth, a society begins to improve its relationship with the environment and levels of environmental degradation reduces.

From a very simplistic viewpoint, it can suggest that economic growth is good for the environment. However, critics argue there is no guarantee that economic growth will lead to an improved environment in fact, the opposite is often the case.

At the least, it requires a very targeted policy and attitudes to make sure that economic growth is compatible with an improving environment.

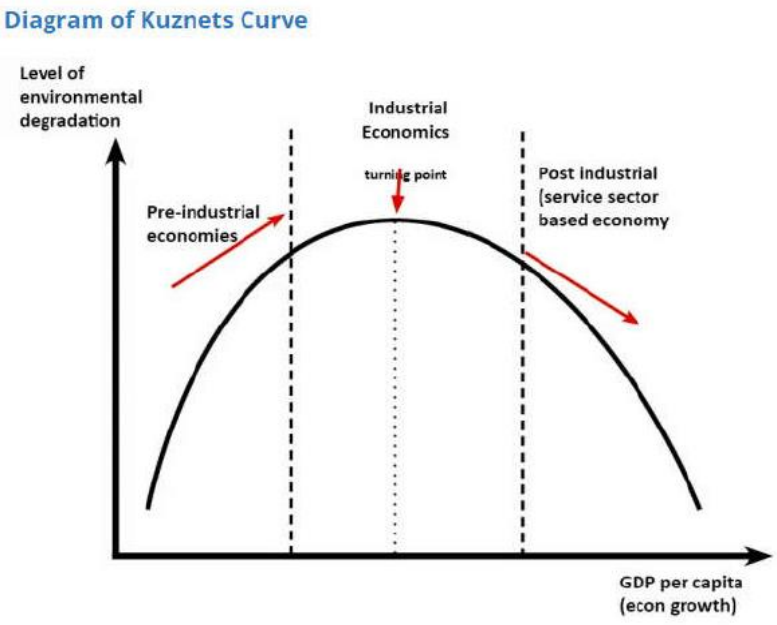


Fig 2.1 Environmental Kuznet Curve

Causes of Environmental Kuznets curve:

1. Empirical evidence of declining pollution levels with economic growth. Studies found that higher economic growth in the US led to increased use of cars, but at the same time due to regulation, levels of air pollution.

2. Spare income with growth. With higher rates of economic growth, people have more discretionary income after paying for basic necessities; therefore, they are more amenable to paying higher prices in return for better environmental standards.

3. Focus on living standards as opposed to real GDP. Traditional economic theory concentrates on increasing real GDP and rates of economic growth. But there is a growing awareness the link between economic growth and living standards can be weak. Focusing on living standards can become politically popular.

4. Improved technology. The primary driving force behind long-term economic growth is improved technology and higher productivity. With higher productivity, we can see higher output, with less raw materials used. For example, since the 1950s, the technology of car use has significantly improved fuel efficiency. In the 1950s, many cars had very low miles per gallon. In recent years, car manufacturers have made strides in reducing fuel consumption and have started to develop hybrid technology.

5. Solar and renewable energy. A good example of how improved technology has reduced potential for environmental damage is the progress in solar technology. In recent years, the cost of solar energy has significantly fallen raising the prospect of clean technology.

6. De-industrialisation. Initially, economic development leads to shifting from farming to manufacturing. This leads to greater environmental degradation. However, increased productivity and rising real incomes see a third shift from industrial to the service sector. An economy like the UK has seen industrialisation shrink as a share of the economy. The service sector usually has a lower environmental impact than manufacturing.

7. Role of government regulation. Economic growth and development usually see a growth in the size of government as a share of GDP. The government are able to implement taxes and regulations in an attempt to solve environmental externalities which harm health and living standards.

8. Diminishing marginal utility of income. Rising income has a diminishing marginal utility. The benefit from your first ₹ 10,00,000 annual income is very high. But, if income rises from ₹90,00,000- ₹1,00,00,000 the gain is very limited in comparison. Having a very high salary is of little consolation if you live with environmental degradation (e.g. congestion, pollution and ill health). Therefore, a rational person who is seeing rising incomes will begin to place greater stress on improving other aspects of living standards.

Criticisms of Kuznets Environmental Curve:

1. Empirical evidence is mixed. There is no guarantee that economic growth will see a decline in pollutants.

2. Pollution is not simply a function of income, but many factors. For example, the effectiveness of government regulation, the development of the economy, population levels.

3. Global pollution. Many developed economies have seen a reduction in industry and growth in the service sector, but they are still importing goods from developing countries. In that sense, they are exporting environmental degradation. Pollution may reduce in the UK, US, but countries who export to these countries are seeing higher levels of environmental degradation. One example is with regard to deforestation. Higher-income countries tend to stop the process of deforestation, but at the same time, they still import meat and furniture
from countries who are creating farmland out of forests.

4. Growth leads to greater resource use. Some economists argue that there is a degree of reduced environmental degradation post industrialisation. But, if the economy continues to expand, then inevitably some resources will continue to be used in greater measure.
There is no guarantee that long-term levels of environmental degradation will continue to fall.

5. Countries with the highest GDP have highest levels of CO2 emission. For example, US has CO2 emissions of 17.564 tonnes per capita. Ethiopia has by comparison 0.075 tonnes per

capita. China's CO_2 emissions have increased from 1,500 million tonnes in 1981 to 8,000 million tonnes in 2009.

6. The shape of the curve may be N-shaped instead of invented U Shaped if the level of environmental degradation after declining for some time again starts increasing as nations incomes continue to increase. Arrow argues that the inverted U-shaped relationship would appear to be false, if pollution increases again at the end due to higher levels of income and mass consumption.

VIII) COMMON RESOURCES

Common resource means a good or service shared by a well-defined community. The community controls the use of such resource by individuals. However, enforcement is weak due to difficulties in monitoring. For example, water in a village pond, which is a common property resource, is used by the villagers only. The village as a community decides upon the manner and the purpose for which the pond water can be used, which results in a set of norms, evolved over time, and largely unwritten. In case of a breach of the norms, however, imposition of penalty is poorly enforced due to poor monitoring, subjectivity in the norms and ambiguities in property rights. The common property regime for managing natural resources is frequently misunderstood. It is often observed as a situation in which there is no management regime in place; as a situation of open access, which is free for all. Accordingly, resource degradation in the developing countries is incorrectly attributed to 'common property systems', whereas it actually originates in the dissolution of local, level institutional arrangements. Therefore, there is a need to properly understand the common property resources and its management systems as these have direct bearing on the sustainable development of natural resources.

We can list a large number of CPRs, which can be brought under the broad headings like land resources, forest resources, water resources, and fishery resources. These resources are being degraded overtime due to overuse or lack of proper management. We shall discuss briefly about these common property resources.

Land Resources:

Common property land resource refers to lands identified with a specific type of property rights. The common lands covered in the National Sample Survey (NSS) enquiry are panchayat lands, government revenue lands, village common lands, village thrashing lands, unclassified forest lands, woodlands and wastelands, river banks, and lands belonging to other households used as commons.

Forest Resources:

Another category of land for which common property rights may exist is land under forests. Unclassified forests, with very low productivity, are always open to use by local communities: Accordingly, both protected and unclassified forests are treated as forming a part of common property forest resources. It is, therefore, the subset of total forest area minus reserve forests to which common property rights are assumed to exist.

Water Resources:

There are a variety of resources of water, which are in the public domain, and a significant part of these are included in the category of commons. Examples are flows of rivers, tanks and natural lakes, groundwater, wetland and mangrove areas, and such other water bodies. Man-made water resources such as dams and canals, tube wells, other wells, and supply of all types of potable water also fall in the category of CPRs depending upon their property rights. Unfortunately, even after many debates about property rights (such as traditional rights, community rights, and basic need human rights), water has not yet been declared as CPR in India, though references are made in the water policy document indirectly. By and large, water resources in India are in common property regimes only. Irrigation canals are managed jointly by the government and communities. Traditionally, tanks, village ponds, and lakes - all of which are treated as CPRs -are sources of water for drinking, livestock rearing, washing, fishing and bathing, and several sanitary-related activities.

IX) TRAGEDY OF COMMON

People have always a tendency to use (misuse) public property according to their whims and fancies. As the public property is not owned by any individual, no one can claim for an exclusive ownership. The net result being misuse of public properties. Perhaps this is the main reason for garbage appearing in the public road, discharging effluents into the river, public parks being misused, public buildings being disfigured etc.

Prof. Garrett Hardin examined the reasons why public properties are either being mis utilised or over utilized by the people. The answer that he identified has been published in the article titled “The Tragedy of commons” (1968). He had studied the character of herdsmen in England. Hardin anxiously watched out the peculiar behaviour of herdsmen that they are always prepared to add additional cattle into the pasture land in England. The logic prevailed that the farmer who grazed the most cattle stood to benefit most from the commons. But the tragedy of this kind of action is that the land was overgrazed and destroyed. This came to be known as “Tragedy of Commons”. Though the tragedy of commons is an observation based on the real

experience example, it finds its applicability in most of the situation in which the resources are owned by the public. There is a tendency to over exploit public resources resulting in total destruction or non-availability of further resources.

X) EXTERNALITIES AND PROPERTY RIGHTS

Externalities:

Externalities occur in an economy when the production or consumption of a specific good or service impacts a third party that is not directly related to the production or consumption of that good or service.

Almost all externalities are considered to be technical externalities. Technical externalities have an impact on the consumption and production opportunities of unrelated third parties, but the price of consumption does not include the externalities. This exclusion creates a gap between the gain or loss of private individuals and the aggregate gain or loss of society as a whole.

The action of an individual or organization often results in positive private gains but detracts from the overall economy. Many economists consider technical externalities to be market deficiencies, and this is the reason people advocate for government intervention to curb negative externalities through taxation and regulation.

Externalities were once the responsibility of local governments and those affected by them. So, for instance, municipalities were responsible for paying for the effects of pollution from a factory in the area while the residents were responsible for their healthcare costs as a result of the pollution. After the late 1990s, governments enacted legislation imposing the cost of externalities on the producer. This legislation increased costs, which many corporations passed on to the consumer, making their goods and services more expensive.

Positive and Negative Externalities:

Most externalities are negative. Pollution is a well-known negative externality. A corporation may decide to cut costs and increase profits by implementing new operations that are more harmful to the environment. The corporation realizes costs in the form of expanding operations but also generates returns that are higher than the costs. However, the externality also increases the aggregate cost to the economy and society making it a negative externality. Externalities are negative when the social costs outweigh the private costs. Some externalities are positive. Positive externalities occur when there is a positive gain on both the private level and social level. Research and development (R&D) conducted by a company can be a positive externality.

R&D increases the private profits of a company but also has the added benefit of increasing the general level of knowledge within a society.

Similarly, the emphasis on education is also a positive externality. Investment in education leads to a smarter and more intelligent workforce. Companies benefit from hiring employees who are educated because they are knowledgeable. This benefits employers because a better-educated workforce requires less investment in employee training and development costs.

Property rights:

The purpose of the property rights approach is to build on and merge with the standard theory of production and exchange in order to obtain an expanded scope of its validity. It argues that the purpose of trade and production is to exchange bundles of rights to do things with goods that are exchanged. Thus, the value of the goods traded increases and the terms of trade improve with increases in the degree of property rights in those goods. It follows that the scope and content of property rights over resources affects the way people behave in a world of scarcity. Since the same resource cannot simultaneously be used to satisfy competing demands, conflicts of interests have to be resolved. Thus, the structure of property rights in a society at some point in time becomes crucial. There is evidence that the allocation of resources is constrained in specific ways by prevailing property rights assignments. What are owned are rights to use resources and not the resource itself. The strength with which rights are owned can be defined by the extent to which an owner's decision about how a resource will be used determines its use.

Property rights are an instrument of society and they derive their significance from the fact that they help individuals form reasonable expectations of their dealings with others. These expectations find expression in a society's laws, customs, and mores. Property rights specify how people may be benefited and harmed and therefore who must pay whom to modify the actions taken by individuals. Thus, property rights have a close relationship with externalities. An externality occurs whenever the activities of one economic agent influences others in ways that are not taken into account by market transactions. Internalizing refers to a process whereby these effects have to be considered by all the

individuals involved in the action. This could be done by a change in the ownership of rights. The concept of property rights can therefore help internalize externalities. However, such a process involves costs that have to be taken into account. Not only is it important to have a structure of property rights, it is equally important to ensure that these rights are enforced. This is where the legal system comes in. One of the purposes of the legal system is to establish the clear delimitation of rights, on the basis of which the transfer of rights can take place through

the market. It should be possible for one user to buy out the rights of the other users to obtain exclusive usage. The use of a piece of land simultaneously for growing wheat and as a parking lot would produce chaos. To avoid this situation, it would be important to create property rights to allow exclusive use of the land.

The advantage of establishing exclusive rights to use a resource when that use does not harm others is easily understood. However, the situation changes when actions may harm others directly. Consider the example of a doctor and a confectioner having shops next to each other. The confectioner's machine disturbs the doctor in her work. The doctor takes the matter to court and the confectioner is made to stop using his machine.

The question is commonly thought of as one in which A inflicts harm on B and the issue is what should be done to restrain A. This is not entirely correct, however. What we are dealing with is a problem of a reciprocal nature: in the process of avoiding harm to B, we harm A. The real question then is: should A be allowed to harm B or should B be allowed to harm A? The problem is how to avoid more serious harm. Another example is straying cattle destroying crops on neighbouring land. If it is inevitable that some cattle will stray, then an increase in the supply of meat can only be obtained at the expense of a decline in the supply of crops. The nature of the choice is not very clear: do we prefer meat or do we prefer crops? It is impossible to know the correct answer without knowing the value of what is obtained and the corresponding opportunity cost.

Once the legal rights of the parties have been established, it is possible to negotiate and modify those arrangements. In the example of the doctor and the confectioner described above, we can have two cases: (1) the property right is given to the doctor and the confectioner stops producing candies, or (2) the property right is given to the confectioner and the doctor can strike a bargain by which the confectioner waives his right.

There are costs associated with any market activity per se. Such costs are termed transactions costs and in some market transactions such costs can be significant. In a world with zero transactions costs, private negotiations or market transactions will always lead to an efficient outcome as long as property rights are well defined and this outcome will be independent of who owns the right. The exact definition of legal rules has only distributional consequences. In reality, however, transaction costs are not zero. When there are costs associated with striking bargains, we have to compare those costs with the potential allocation gains from striking a bargain. Only in situations where those gains exceed the necessary bargaining costs will Coase-type results hold.

When bargaining costs are high, externalities will distort the allocation of resources and the assignment of property rights can have a major effect on that allocation. If, for example, major industries are given the right to spew noxious fumes into the atmosphere, an efficient allocation is unlikely to emerge since the costs of bringing together into an effective bargaining unit all the individuals harmed by such fumes are probably quite high. Nevertheless, development of the Coase Theorem and later research based on it has had a significant impact on the way economists think of the relationship between externalities, property rights, and the efficient allocation of resources.

The concept of property rights can be extended to various fields. One could think of the problem faced by a financially weak, independent inventor when selling a valuable but easily imitated invention for which no property rights exist. Most independent inventors cannot successfully create an organization to take commercial advantage of their invention, so they have to rely on another party. This involves a production contract, a licensing arrangement, or the outright sale of the invention. Thus, the inventor's ability to make profits of capture rents depends on the market value of the invention, property rights, and the information of the inventor. When the inventor can rely on patents or other mechanism to protect her intellectual property rights, then theory suggests that the inventor can appropriate a substantial fraction of the value of the invention. But if property rights are weak and non-existent, then the inventor's ability to capture rents are limited. Also, reliance on laws concerning theft of ideas does not work since buyers employ strategies to avoid legal challenges. Finally, some forms of intellectual property rights such as new product concepts and management ideas are inherently difficult to protect.

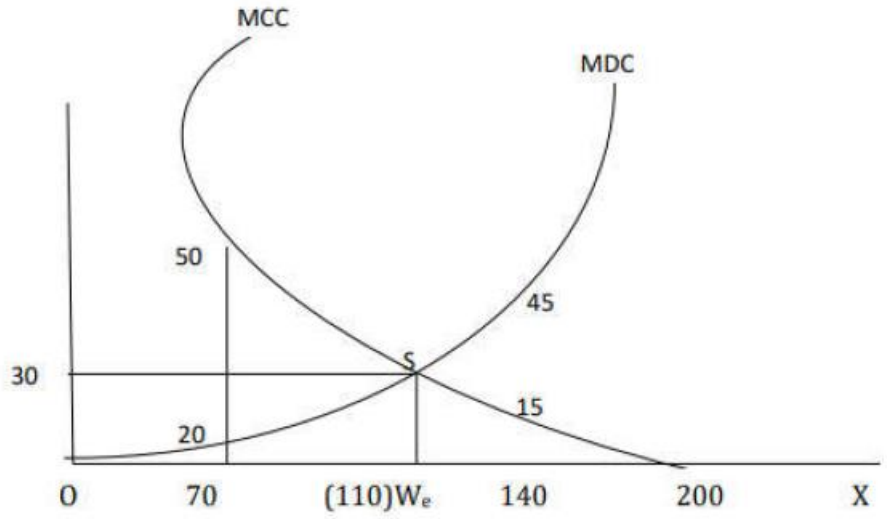
XI) COASE THEOREM

According to Coase, any effort to internalize environmental externalities requires an effective scheme for assigning property rights.

Coase also believed that by assigning property rights to at least one of the parties involved (either polluter or the pollute) there would be no effect on the final outcome of the environmental problem. The Coase theorem developed by economist Ronald Coase in 1960. The advantages of this theory are that the pollution problem can be solved by an arbitrary assignment of property rights. Optimal level of pollution can be attained through voluntary negotiations of private parties (polluter and pollute). If the state is acting as a regulator, enforceable ownership rights have to be assigned so that it can act as private enterprise.

To illustrate the essence of this theory, let us follow the example given below. The two familiar firms to be taken to explain this theory are the paper mill and the fish hatchery. River flowing

nearby these firm are a common good. The fish hatchery believes that and as per the legal rights, the river can be used for its activities. The paper mill is not permitted to discharge the effluents into this river. In the figure, this situation is represented by the origin O, where the amount of waste released into this river from the paper mill is zero.



The figure explains that if the paper mill is not permitted to dispose of the wastes, it has to find an alternative method of disposal of 200 units of waste. But this system cannot sustain for a long period. When the waste discharged by the firm is less than We (110 units). The Marginal Cost of Cleaning (MCC) of the Mill is higher than the Marginal Damage Cost
(MDC) to the hatchery. As shown in the figure, for the 70th unit of waste emitted into the river, the MDC to hatchery is Rs. 20. The MCC of mill is Rs. 50. The Rs. 50 is for cleaning up of 130 units (200-70=130) of wastes. To discharge 70 units of wastes, the mill is prepared to pay Rs. 20 to compensate for the damage caused to hatchery, because the alternative cost for waste disposal of 70 units is Rs. 50. This proposal is advantageous to the paper mill. Though both the parties, the paper mill and the fish hatchery, enjoy some advantages, they can think of a better bargain. These two firms will be in a position to engage in a mutually beneficial transaction provided that it is at the point where the MCC>MDC. Further the negotiation ends when the MCC=MDC. This is the condition for the optimum level of pollution. In figure this is attained at We or 110 units of waste emission.

Coase theorem goes beyond the mere attainment of optimally. It also states that this optimal outcome is completely independent of the two parties who have the rights to the river. To illustrate this, let us imagine that the paper mill has exclusive legal rights to the use of the river. Under this circumstance, the mill can dispose of the entire wastes to the river. The figure shows that the paper mill can discharge a total of 200 units. But for each units between 110 and 200 units of wastes discharged, the MDC is greater than MCC. It means that the mill's MCC for abating pollution is lower than MDC which the mill needs to meet. This situation will call for the two firms. The paper mill and the hatchery to engage in a mutually beneficial transaction. When the waste is 140 units, the control cost which the mill needs to pay to the hatchery is Rs. 15 per unit, whereas the mill itself needs to spend Rs. 45 to avoid the 1 unit of waste emission. Thus, when the emission level is at 140 units, the MDC is greater than MCC. The hatchery will take initiative to offer any amount higher than Rs. 15 to avoid higher levels of pollution emission to the river. Thus, the hatchery moves on the MCC curve and finally, it settles at S where MDC=MCC. Thus, the optional level of pollution is again reached at We or 110 units where MDC=MCC.

The theory based on several assumptions:

a. Every firm has perfect information

b. Consumer and producers are price takers

c. Producers maximize the profit and consumers maximize the utility

d. There are no income or wealth effects e. There are no transaction costs.

Critical evaluation:

The following major limitations of this theory are:

1. Wealth effect is assumed to be non-existent. But in reality, we all of us know that there are wealth effects which are subject to environmental factors.

2. Complete set of property rights is necessary to obtain optimum allocation of resources. Coase says that for achieving efficiency it does not matter how these rights are distributed. The question who will assign property rights to public goods is still a hard nut to crack.

Arbitrary valuation cannot be considered as the relatively better option.

3. The transaction costs will be much higher when the parties involved in the negotiation process are many.

4. Coase theorem appears to be indifferent from the polluter pay principal which states that it is the polluter who has to meet the environmental damage cost. The extent of optimality in the polluter pay principle is analysed in the Pigouvian fee.